

Empirical Methods in Finance

Class 11 – Dynamic Correlation Models, ERC and Empirical Betas

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Objectives of the lecture

This class has four goals:

1. understand why correlations are not constant when financial markets become agitated,
2. introduce multivariate GARCH models for conditional covariance matrices,
3. discuss dynamic correlation models and non-Gaussian dependence,
4. connect the theory to portfolio construction through Equal Risk Contribution portfolios and DCC-implied empirical betas.

Portfolio motivation: ERC beyond two assets

Why ERC is a useful entry point

Equal Risk Contribution (ERC) is a portfolio construction rule. It is also a simple way to understand why conditional covariance matrices matter.

With more than two assets, the allocation depends on a joint risk measure:

$$\Sigma_t = D_t^{1/2} \Gamma_t D_t^{1/2}. \quad (1)$$

This matrix combines two objects:

- individual conditional variances in D_t ,
- conditional correlations in Γ_t .

The message is direct: changing correlations can change optimal weights, even when individual volatilities are unchanged.

Risk contribution in a multi-asset portfolio

Let w_t be an $n \times 1$ vector of portfolio weights. Portfolio volatility is:

$$\sigma_{p,t} = \sqrt{w_t' \Sigma_t w_t}. \quad (2)$$

The marginal contribution of asset i to portfolio volatility is:

$$\frac{\partial \sigma_{p,t}}{\partial w_{i,t}} = \frac{(\Sigma_t w_t)_i}{\sigma_{p,t}}. \quad (3)$$

Its total risk contribution is:

$$RC_{i,t} = w_{i,t} \frac{(\Sigma_t w_t)_i}{\sigma_{p,t}}. \quad (4)$$

ERC chooses weights such that:

$$RC_{1,t} = RC_{2,t} = \dots = RC_{n,t}. \quad (5)$$

The interaction between volatility and correlation

For two assets, intuition is easy. For many assets, it becomes matrix-based.

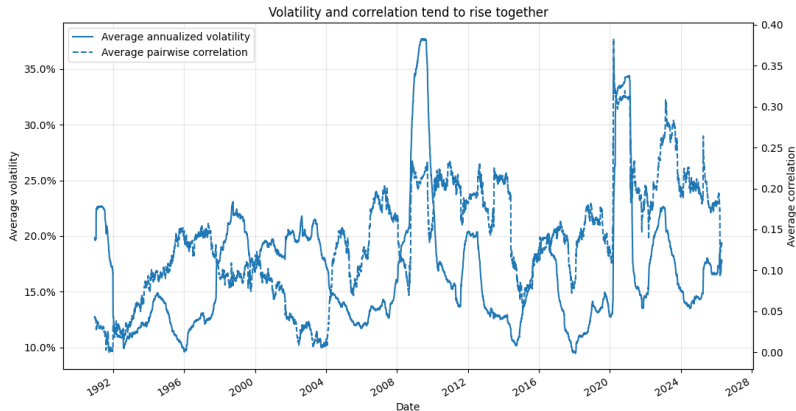
A low-volatility asset can still contribute a lot to portfolio risk if it is highly correlated with the other positions.

A high-volatility asset can be useful if its correlation profile is diversifying.

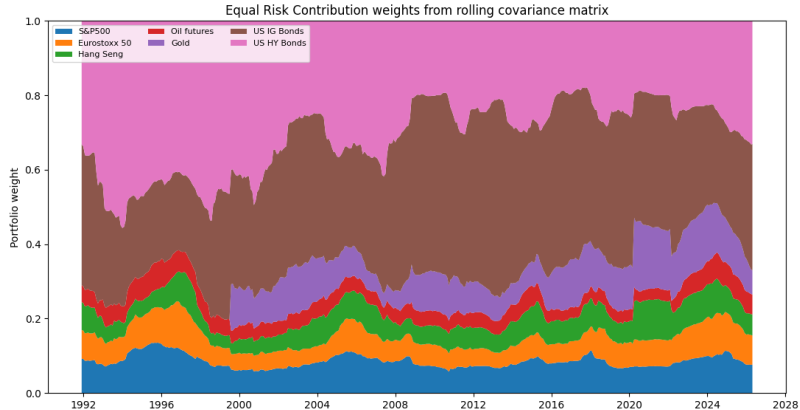
This is why ERC is not an inverse-volatility portfolio. It is an allocation based on the full covariance matrix:

$$w_t^{ERC} = \arg \min_{w_t} \sum_{i=1}^n \left(\frac{RC_{i,t}}{\sigma_{p,t}} - \frac{1}{n} \right)^2, \quad \sum_{i=1}^n w_{i,t} = 1, \quad w_{i,t} \geq 0. \quad (6)$$

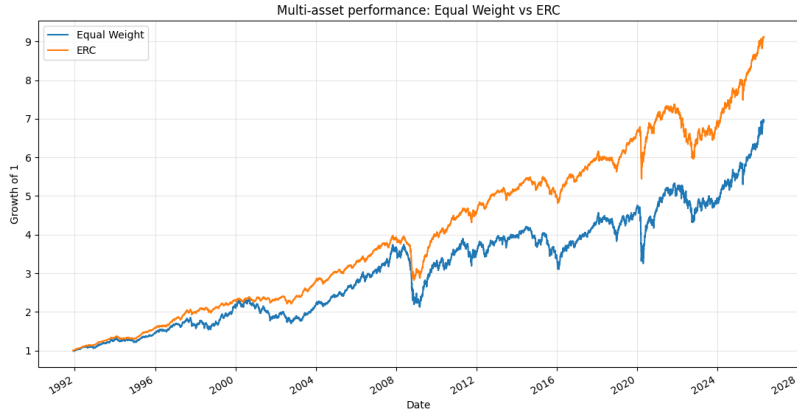
Empirical motivation: volatility and correlation move together



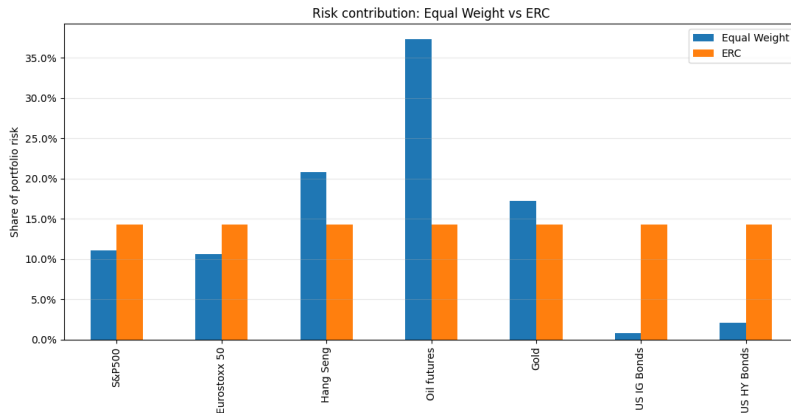
ERC weights react to the covariance matrix



Why this matters for performance and drawdowns



Same weights, different risk contributions



A simple empirical result from the class dataset

The empirical illustration uses daily prices from 1990 to April 2026 for seven assets: US equities, European equities, Hong Kong equities, oil, gold, US investment-grade bonds and US high-yield bonds.

Missing observations are carried forward before computing log returns. Since log returns require positive prices, non-positive prices are treated as missing and carried forward.

Portfolio	Ann. return	Ann. vol	Sharpe	Max drawdown
Equal Weight	5.52%	10.34%	0.53	-43.13%
ERC	6.32%	5.29%	1.19	-29.02%

This is not a trading recommendation. It is a classroom example showing that covariance estimation affects allocation, risk concentration and realized drawdowns.

Dynamic correlation models

Objectives of the lecture

Main characteristic of the multivariate distribution: the **dependency parameter**. It measures the strength of the link between two series.

For a number of standard distributions (namely, the elliptical family, which includes the normal and the Student t distributions), dependency is simply measured by the **Pearson's (or linear) correlation coefficient**. For this reason, we often refer to the concept of **correlation** instead of **dependency**. How does correlation between market returns vary when **markets become agitated**? This is a crucial issue from an asset-management perspective.

Objectives of the lecture

Most **asset allocations** are based on the use of a correlation matrix computed over a given sample period. If correlation increases during turbulent periods, the benefits of diversification disappear when they are the most needed, i.e. during crashes.

Important concept: Tail dependence, that is the joint realization of extreme losses. It could be generated by

1. a dynamic correlation model
2. a distribution with different levels of dependence.

Objectives of the lecture

How to test the constancy of the dependency parameter?

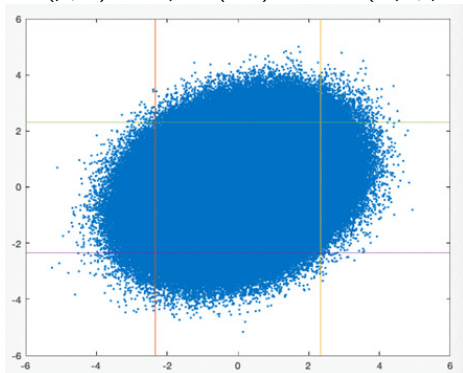
- **Test the equality of linear correlation coefficients** computed before and after a crash. This approach may however be misleading because conditioning the estimation of the correlation coefficient on the sample period induces an estimator bias, if the variance changes over the 2 subperiods.
- **Test in a conditional model:**
 1. Estimate the joint dynamics of stock-market returns and
 2. Describe how the conditional correlation varies over time.

We need to model the joint dynamics of a number of series:

- Multivariate GARCH models (for the covariance matrix),
- Multivariate distributions or copulas models (for the distribution).

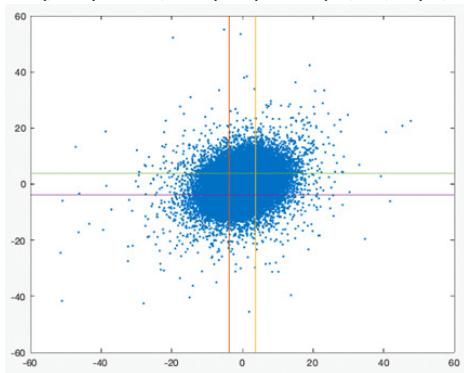
Objectives of the lecture

Simulation of a $N(\mu, \Sigma)$ with $\mu = (0; 0)$ and $\Sigma = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$ and $\rho = 0.25$



Objectives of the lecture

Simulation of a $t(\mu, \Sigma)$ with $\mu = (0; 0)$, $\Sigma = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$, $\rho = 0.25$, $\nu = 4.1$



Correlation below 1% quantile = 44%

Objectives of the lecture

Outline of the class:

1. Portfolio motivation: Equal Risk Contribution and covariance matrices
2. Multivariate GARCH models
3. Estimation issues
4. Dynamic conditional correlation
5. Non-normal distributions and copulas
6. Empirical application using the multi-asset dataset

Multivariate GARCH models

We consider a vector of returns $r_t = (r_{1,t}, \dots, r_{n,t})'$ with joint dynamics

$$r_t = \mu_t(\theta) + \varepsilon_t \quad (7)$$

$$\text{with } \mu_t(\theta) = E[r_t | I_{t-1}] = \mu(\theta, I_{t-1}) \quad (8)$$

$$\varepsilon_t = \Sigma_t^{1/2}(\theta) z_t \quad (9)$$

$$\text{with } \Sigma_t(\theta) = E[(r_t - \mu_t(\theta))(r_t - \mu_t(\theta))' | I_{t-1}] = \Sigma(\theta, I_{t-1}). \quad (10)$$

where $\mu_t(\theta)$ denotes the $(n \times 1)$ vector of conditional means, $\Sigma_t(\theta)$ denotes the $(n \times n)$ conditional covariance matrix of the error term ε_t and θ the vector of unknown parameters.

The standardized innovation vector $z_t = \Sigma_t^{-1/2}(\theta)\varepsilon_t$ is iid with mean $E[z_t] = 0$ and covariance matrix $V[z_t] = I_n$.

We assume that z_t is drawn from the multivariate normal $N(0, I_n)$ distribution.

Multivariate GARCH models

The multivariate normal distribution

The random vector $X = (X_1, \dots, X_n)$ has a multivariate normal distribution with zero mean and identity covariance matrix, denoted $X \sim N_n(0, I_n)$, if its distribution is

$$\varphi(x) = \frac{1}{(2\pi)^{n/2}} \exp\left(-\frac{1}{2}x'x\right) \quad (11)$$

With the normal distribution, dependency is naturally controled through the covariance matrix.

The random vector $Z = (Z_1, \dots, Z_n)$ has a multivariate normal distribution with $(n \times 1)$ mean vector μ and $(n \times n)$ covariance matrix Σ , denoted $Z \sim N_n(\mu, \Sigma)$, if

$$Z = \mu + AX \text{ with } \Sigma = AA' \quad (12)$$

so that

$$f(z) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(z - \mu)' \Sigma^{-1} (z - \mu)\right) \quad (13)$$

The **square root of a covariance matrix**

What is the meaning of $\Sigma_t^{1/2}$?

It is a matrix A_t such that $A_t A_t' = \Sigma_t$.

Two possibilities:

- **Cholesky decomposition**, where A_t is lower triangular with positive diagonal entries.
- **Spectral (or Schur) decomposition**, where $\Sigma_t = V_t \Lambda_t V_t'$ and $A_t = V_t \Lambda_t^{1/2} V_t'$, with V_t the matrix of eigenvectors and Λ_t the diagonal matrix of eigenvalues.

Recommendation: The Cholesky decomposition is more appropriate when there is a “natural” ranking of the assets (most important ones first). In the other cases, the spectral decomposition is probably the safest approach.

Several parameterizations have been proposed for $\Sigma_t(\theta)$.

The main issue is the **dimensionality of the parameter vector** when the number of variables n increases.

It globally is a trade-off between:

- capturing the main statistical features of the distribution
- estimating a large number of parameters
- incorporating additional constraints such that the covariance matrix remains positive definite at each date t .

Two subsidiary choices:

- Should the conditional correlation or conditional covariance be modelled?
- Should correlation or covariance (or both) be dynamic?

Multivariate GARCH models – Notations

Define D_t the $(n \times n)$ diagonal matrix with the conditional variances

$$D_t = \begin{pmatrix} \sigma_{1,t}^2 & 0 & \cdots & 0 \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & \sigma_{n,t}^2 \end{pmatrix} \quad (14)$$

Also define Γ_t the $(n \times n)$ matrix of conditional correlations of ε_t as

$$\Gamma_t = D_t^{-1/2} \Sigma_t D_t^{-1/2} = \{\rho_t\}_{ij}. \quad (15)$$

We deduce the $(n \times 1)$ vector of normalized innovations $u_t = D_t^{-1/2} \varepsilon_t$.

u_t differs from the standardized innovation $z_t = \Sigma_t^{-1/2} \varepsilon_t$, since it not orthogonalized.

Multivariate GARCH models – Vech GARCH

Kraft and Engle (1982, *WP*), Bollerslev, Chou, and Kroner (1988, *J. Econometrics*).

Each element of the covariance matrix is a linear function of the most recent past cross-products of errors and conditional variances and covariances.

The **Vech GARCH(1,1) model** is defined as

$$\text{vech}(\Sigma_t) = \text{vech}(\Omega) + A \text{vech}(\varepsilon_{t-1}\varepsilon'_{t-1}) + B \text{vech}(\Sigma_{t-1}) \quad (16)$$

where:

- Ω is an $(n \times n)$ positive definite and symmetric matrix,
- A and B are $(n(n+1)/2 \times n(n+1)/2)$ matrices;
- *vech* (for *vector half*) is the operator which stacks the lower triangular elements of an $(n \times n)$ matrix as an $(n(n+1)/2 \times 1)$ vector.

Multivariate GARCH models – Vech GARCH

In the case $n = 2$, it can be written as

$$\begin{pmatrix} \sigma_{1,t}^2 \\ \sigma_{12,t} \\ \sigma_{2,t}^2 \end{pmatrix} = \begin{pmatrix} \omega_{11} \\ \omega_{12} \\ \omega_{22} \end{pmatrix} + \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} \varepsilon_{1,t-1}^2 \\ \varepsilon_{1,t-1}\varepsilon_{2,t-1} \\ \varepsilon_{2,t-1}^2 \end{pmatrix} \quad (17)$$

$$+ \begin{pmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{pmatrix} \begin{pmatrix} \sigma_{1,t-1}^2 \\ \sigma_{12,t-1} \\ \sigma_{2,t-1}^2 \end{pmatrix} \quad (18)$$

The number of unknown parameters is $[n(n+1)/2][1+2n(n+1)/2]$ (21 for $n = 2$).

Although this specification is very flexible, the **number of parameters is proportional to n^4** and therefore increases very rapidly with the number of variables.

In addition, conditions which ensure that the conditional covariance matrices are positive definite are difficult to verify and impose.

Multivariate GARCH models – Diagonal vec GARCH

Bollerslev, Engle, and Wooldridge (1988, *JPE*).

In the **Diagonal vec GARCH model**, each element of the covariance matrix ($\sigma_{ij,t}$) only depends on the corresponding past elements $\sigma_{ij,t-1}$ and $\varepsilon_{i,t-1}\varepsilon_{j,t-1}$:

$$\Sigma_t = \Omega^* + A^* \circ (\varepsilon_{t-1}\varepsilon'_{t-1}) + B^* \circ \Sigma_{t-1} \quad (19)$$

where the matrix Ω^* is an $(n \times n)$ positive definite and symmetric, the matrices A^* and B^* are $(n \times n)$ positive semidefinite and symmetric matrices, and $\circ$ denotes the Hadamard product (i.e., $\{A \circ B\}_{ij} = A_{ij}B_{ij}$).

The variances ($\sigma_{1,t}^2$ and $\sigma_{2,t}^2$) follow univariate GARCH(1,1) processes. The covariance $\sigma_{12,t}$ has a similar structure.

Imposing that Σ_t is positive definite can be done by parameterizing the model with Cholesky matrices:

$$\Omega^* = (\Omega^*)^{1/2}(\Omega^*)^{1/2}A^* = (A^*)^{1/2}(A^*)^{1/2}B^* = (B^*)^{1/2}(B^*)^{1/2} \quad (20)$$

Multivariate GARCH models – Diagonal vec GARCH

In the case $n = 2$, it can be written as

$$\begin{pmatrix} \sigma_{1,t}^2 & \sigma_{12,t} \\ \sigma_{12,t} & \sigma_{2,t}^2 \end{pmatrix} = \begin{pmatrix} \omega_{11} & \omega_{12} \\ \omega_{12} & \omega_{22} \end{pmatrix} + \begin{pmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{pmatrix} \circ \begin{pmatrix} \varepsilon_{1,t-1}^2 & \varepsilon_{1,t-1}\varepsilon_{2,t-1} \\ \varepsilon_{1,t-1}\varepsilon_{2,t-1} & \varepsilon_{2,t-1}^2 \end{pmatrix} \quad (21)$$

$$+ \begin{pmatrix} b_{11} & b_{12} \\ b_{12} & b_{22} \end{pmatrix} \circ \begin{pmatrix} \sigma_{1,t-1}^2 & \sigma_{12,t-1} \\ \sigma_{12,t-1} & \sigma_{2,t-1}^2 \end{pmatrix} \quad (22)$$

$$= \begin{pmatrix} \omega_{11} & \omega_{12} \\ \omega_{12} & \omega_{22} \end{pmatrix} + \begin{pmatrix} a_{11}\varepsilon_{1,t-1}^2 & a_{12}\varepsilon_{1,t-1}\varepsilon_{2,t-1} \\ a_{12}\varepsilon_{1,t-1}\varepsilon_{2,t-1} & a_{22}\varepsilon_{2,t-1}^2 \end{pmatrix} \quad (23)$$

$$+ \begin{pmatrix} b_{11}\sigma_{1,t-1}^2 & b_{12}\sigma_{12,t-1} \\ b_{12}\sigma_{12,t-1} & b_{22}\sigma_{2,t-1}^2 \end{pmatrix} \quad (24)$$

This model reduces the number of unknown parameters considerably to $3[n(n+1)/2]$, so that it is proportional to n^2 only (9 for $n = 2$).

Multivariate GARCH models – BEKK model

Engle and Kroner (1995, *Econometric Theory*).

$$\Sigma_t = \tilde{\Omega} + \tilde{A}'(\varepsilon_{t-1}\varepsilon'_{t-1})\tilde{A} + \tilde{B}'\Sigma_{t-1}\tilde{B} \quad (25)$$

where $\tilde{\Omega}$, $\tilde{A}$ and $\tilde{B}$ are $(n \times n)$ matrices.

The main advantage of this **BEKK specification** is that the conditional covariance matrix is positive definite as long as $\tilde{\Omega}$ is positive definite.

In the case $n = 2$, it can be written as

$$\begin{pmatrix} \sigma_{1,t}^2 & \sigma_{12,t} \\ \sigma_{12,t} & \sigma_{2,t}^2 \end{pmatrix} = \begin{pmatrix} \omega_{11} & \omega_{12} \\ \omega_{12} & \omega_{22} \end{pmatrix} + \begin{pmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \end{pmatrix} \begin{pmatrix} \varepsilon_{1,t-1}^2 & \varepsilon_{1,t-1}\varepsilon_{2,t-1} \\ \varepsilon_{1,t-1}\varepsilon_{2,t-1} & \varepsilon_{2,t-1}^2 \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \quad (26)$$

$$+ \begin{pmatrix} b_{11} & b_{21} \\ b_{12} & b_{22} \end{pmatrix} \begin{pmatrix} \sigma_{1,t-1}^2 & \sigma_{12,t-1} \\ \sigma_{12,t-1} & \sigma_{2,t-1}^2 \end{pmatrix} \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} \quad (27)$$

This model involves $[n(n+1)/2] + 2n^2$ unknown parameters (11 for $n = 2$).

Multivariate GARCH models – BEKK model - Special Cases

Diagonal BEKK model

$$\begin{pmatrix} \sigma_{1,t}^2 & \sigma_{12,t} \\ \sigma_{12,t} & \sigma_{2,t}^2 \end{pmatrix} = \begin{pmatrix} \omega_{11} & \omega_{12} \\ \omega_{12} & \omega_{22} \end{pmatrix} + \begin{pmatrix} a_{11} & 0 \\ 0 & a_{22} \end{pmatrix} \begin{pmatrix} \varepsilon_{1,t-1}^2 & \varepsilon_{1,t-1}\varepsilon_{2,t-1} \\ \varepsilon_{1,t-1}\varepsilon_{2,t-1} & \varepsilon_{2,t-1}^2 \end{pmatrix} \begin{pmatrix} a_{11} & 0 \\ 0 & a_{22} \end{pmatrix} \quad (28)$$

$$+ \begin{pmatrix} b_{11} & 0 \\ 0 & b_{22} \end{pmatrix} \begin{pmatrix} \sigma_{1,t-1}^2 & \sigma_{12,t-1} \\ \sigma_{12,t-1} & \sigma_{2,t-1}^2 \end{pmatrix} \begin{pmatrix} b_{11} & 0 \\ 0 & b_{22} \end{pmatrix} \quad (29)$$

$\Rightarrow$ (7 unknown parameters).

Scalar BEKK model

$$\begin{pmatrix} \sigma_{1,t}^2 & \sigma_{12,t} \\ \sigma_{12,t} & \sigma_{2,t}^2 \end{pmatrix} = \begin{pmatrix} \omega_{11} & \omega_{12} \\ \omega_{12} & \omega_{22} \end{pmatrix} + a^2 \begin{pmatrix} \varepsilon_{1,t-1}^2 & \varepsilon_{1,t-1}\varepsilon_{2,t-1} \\ \varepsilon_{1,t-1}\varepsilon_{2,t-1} & \varepsilon_{2,t-1}^2 \end{pmatrix} + b^2 \begin{pmatrix} \sigma_{1,t-1}^2 & \sigma_{12,t-1} \\ \sigma_{12,t-1} & \sigma_{2,t-1}^2 \end{pmatrix} \quad (30)$$

$\Rightarrow$ (5 unknown parameters).

Multivariate GARCH models – The constant term

Engle and Mezrich (1996, *Risk*).

All the models can be estimated with the additional constraint that **the long-run covariance matrix is equal to the sample covariance matrix**.

This constraint reduces the number of parameters dramatically and often improves the finite-sample performance.

We have the following parametrization:

- $\text{vech}(\Omega) = (I_{n(n+1)/2} - A - B) \text{vech}(S)$ (vech model)
- $\Omega^* = (ee' - A^* - B^*) \circ S$ (diagonal vec model)
- $\tilde{\Omega} = S - \tilde{A} S \tilde{A}' - \tilde{B} S \tilde{B}'$ (BEKK model)

where $S = \frac{1}{T} \sum_{t=1}^T \hat{\varepsilon}_t \hat{\varepsilon}_t'$ is the sample covariance matrix of residuals, and $e = (1, \dots, 1)'$.

Problems with first-generation multivariate GARCH models:

- **Problem of dimensionality:** difficult to handle with large-dimensional systems.
⇒ **Solutions:** Factor GARCH models (Engle, Ng, and Rothschild, 1990) and Flexible GARCH model (Ledoit, Santa-Clara, and Wolf, 2003).
- **Restrictions for positive definite matrix:** often difficult to impose, except Diagonal Vec and BEKK.
- **Dynamics of the correlation:** describe the dynamics of covariances but not of correlations, which we are mostly interested in.
⇒ **Solutions:** CCC GARCH models (Bollerslev, 1990) and DCC GARCH models (Engle and Sheppard, 2001).

Multivariate GARCH models – Factor GARCH model

Engle, Ng, and Rothschild (1990, *J. Econometrics*), Bollerslev and Engle (1993, *Econometrica*): **Factor models**.

This model has been introduced to further reduce the **number of unknown parameters**.

The idea of this parametrization is that the joint dynamics of the $(n \times 1)$ vector of returns r_t can be described using a small number of observed factors K ($K < n$).

The description of r_t is given by the $(K \times 1)$ vector of factors f_t and the $(n \times K)$ matrix B of time-invariant factor loadings:

$$r_t = B f_t + \varepsilon_t \quad (31)$$

where we assume that the error term ε_t has constant conditional covariance matrix Ψ , that the factors f_t have conditional covariance matrix Λ_t and that ε_t and f_t are uncorrelated.

Multivariate GARCH models – Factor GARCH model

The conditional covariance matrix of r_t is equal to

$$V_{t-1}(r_t) = \Sigma_t = \Psi + B\Lambda_t B'. \quad (32)$$

Assume now that the conditional covariance matrix of factors Λ_t is diagonal with elements $\lambda_{k,t}$ given by univariate GARCH(1,1) model, then the model simplifies as

$$\Sigma_t = \Psi + \sum_{k=1}^K \beta_k \beta_k' \lambda_{k,t}, \quad (33)$$

where β_k denotes the k -th column in B .

In the very simple case $n = 2$ (with $K = 1$ factor), it can be written as

$$\begin{pmatrix} \sigma_{1,t}^2 & \sigma_{12,t} \\ \sigma_{12,t} & \sigma_{2,t}^2 \end{pmatrix} = \begin{pmatrix} \omega_{11} & \omega_{12} \\ \omega_{12} & \omega_{22} \end{pmatrix} + \begin{pmatrix} \beta_1^2 \lambda_t & \beta_1 \beta_2 \lambda_t \\ \beta_1 \beta_2 \lambda_t & \beta_2^2 \lambda_t \end{pmatrix} \quad (34)$$

Remark: It is also possible to define a Principal Component GARCH model, in which the factors are determined statistically.

Multivariate GARCH models – CCC model

Bollerslev (1990): **Constant conditional correlation model.**

The time-varying conditional covariances are parametrized to be **proportional to the product of the corresponding conditional standard deviations.**

Assume that each conditional variance is a univariate GARCH(1,1) model as

$$\sigma_{i,t}^2 = \omega_i + \alpha_i \varepsilon_{i,t-1}^2 + \beta_i \sigma_{i,t-1}^2 \quad (35)$$

where ω_i , α_i , and β_i are non-negative and $\alpha_i + \beta_i < 1$ for all $i = 1, \dots, n$.

We then assume that the conditional correlation matrix $\Gamma_t = \Gamma = \{\rho\}_{ij}$ is a time-invariant $(n \times n)$ positive definite parameter matrix with unit diagonal elements

$$\Gamma = \begin{pmatrix} 1 & \rho_{12} & \cdots & \rho_{1n} \\ \rho_{12} & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \rho_{n-1,n} \\ \rho_{1n} & \cdots & \rho_{n-1,n} & 1 \end{pmatrix} \quad (36)$$

The temporal variation in Σ_t is determined solely by the conditional variances

$$\Sigma_t = D_t^{1/2} \Gamma D_t^{1/2} \quad (37)$$

The matrix Σ_t is almost surely positive definite for all t . In addition, it is covariance stationary iif $\alpha_i + \beta_i < 1$ for all $i = 1, \dots, n$. Finally, since ε_t has finite, time-independent second moments $\sigma_{ij} = E[\sigma_{ij,t}]$, it is also a covariance stationary process.

This approach decreases the number of parameters dramatically: We only need to estimate the dynamics of the n conditional variances and the constant correlation matrix, so that the number of parameters reduces to $3n + n(n+1)/2$ (9 for $n = 2$).

In addition, as presented before, the correlation matrix Γ may be estimated in a preliminary step using the sample correlation matrix of residuals.

Remark: the CCC model is a useful starting point for multivariate modeling, but the constancy of conditional correlation is an unrealistic feature.

Multivariate GARCH models – DCC model

Engle and Sheppard (2001): **Dynamic conditional correlation model.**

The basic idea is that the conditional correlation matrix Γ_t is in fact time varying, so that the conditional covariance matrix is now

$$\Sigma_t = D_t^{1/2} \Gamma_t D_t^{1/2} \quad (38)$$

The conditional correlation matrix is defined by:

$$\Gamma_t = \text{diag}(Q_t)^{-1/2} \times Q_t \times \text{diag}(Q_t)^{-1/2} \quad (39)$$

$$Q_t = (1 - \alpha - \beta) \bar{Q} + \alpha(u_{t-1} u'_{t-1}) + \beta Q_{t-1} \quad (40)$$

$$u_t = D_t^{-1/2} \varepsilon_t = \{\varepsilon_{i,t} / \sigma_{i,t}\}_{i=1, \dots, n} \quad (41)$$

where $\bar{Q}$ is the sample covariance matrix of u_t and α and β satisfy $0 \leq \alpha, \beta \leq 1$ and $\alpha + \beta \leq 1$. Once these restrictions are imposed, the conditional correlation matrix (Γ_t) is guaranteed to be positive definite. The $\text{diag}(Q_t)$ of an $(n \times n)$ matrix is the $(n \times 1)$ vector containing the diagonal of Q_t .

Multivariate GARCH models are estimated in a very similar way as univariate models.

Suppose we have an $(n \times 1)$ vector of time series $\{r_t\}_{t=1}^T$. Unknown parameters are θ . Under the assumption of conditional multivariate normal distribution, the **log-likelihood function** for r_t is

$$\log(L_T(\theta)) = \sum_{t=1}^T \log(\ell_t(\theta)), \quad (42)$$

where

$$\log(\ell_t(\theta)) = -\frac{n}{2} \log(2\pi) - \frac{1}{2} \left[\log |\Sigma_t(\theta)| + (r_t - \mu_t(\theta))' \Sigma_t^{-1}(\theta) (r_t - \mu_t(\theta)) \right]. \quad (43)$$

The ML estimator $\hat{\theta}_{ML}$ satisfies $L_T(\hat{\theta}_{ML}) \geq L_T(\theta)$, $\forall \theta \in \Theta$, where Θ is the domain of definition of θ .

Multivariate GARCH models – Estimation Issues

$\hat{\theta}_{ML}$ is asymptotically normal with distribution

$$\sqrt{T}(\hat{\theta}_{ML} - \theta_0) \stackrel{d}{\sim} N(0, A_0^{-1}), \quad (44)$$

where A_0 is the information matrix evaluated at the true parameter θ_0 :

$$A_0 = -\frac{1}{T} \sum_{t=1}^T E \left[\frac{\partial^2 \log(\ell_t(\theta_0))}{\partial \theta \partial \theta'} \right]. \quad (45)$$

The inverse of this matrix is less than the asymptotic covariance matrix for all other estimators by a positive definite matrix. In finite sample, this expression is estimated by the Hessian matrix of $\log(L_T(\theta))$ computed at $\hat{\theta}_{ML}$:

$$\hat{A}_0 = -\frac{1}{T} \sum_{t=1}^T \frac{\partial^2 \log(\ell_t(\hat{\theta}_{ML}))}{\partial \theta \partial \theta'}. \quad (46)$$

When the distribution is correctly specified, the information matrix A_0 is shown to be equal to the outer product of the gradients

$$B_0 = \frac{1}{T} \sum_{t=1}^T E \left[\frac{\partial \log(\ell_t(\theta_0))}{\partial \theta} \frac{\partial \log(\ell_t(\theta_0))'}{\partial \theta} \right]. \quad (47)$$

This result has important practical implications, because estimating the covariance matrix using the outer product of the gradients only requires first-order derivatives

$$\hat{B}_T = \frac{1}{T} \sum_{t=1}^T \left[\frac{\partial \log(\ell_t(\hat{\theta}_{ML}))}{\partial \theta} \frac{\partial \log(\ell_t(\hat{\theta}_{ML}))'}{\partial \theta} \right], \quad (48)$$

which are usually more stable than second-order derivatives.

Multivariate GARCH models – Estimation of DCC model

A **two-step estimation** is based on the idea that parameters of the conditional variances (θ_V) and the conditional correlations (θ_C) can be estimated separately.

A justification is that the log-likelihood, with $\theta = (\theta'_V, \theta'_C)'$:

$$\log(L_T(\theta)) = \sum_{t=1}^T \log(\ell_t(\theta)) = -\frac{1}{2} \sum_{t=1}^T \left[n \log(2\pi) + \log |\Sigma_t| + (r_t - \mu_t)' \Sigma_t^{-1} (r_t - \mu_t) \right], \quad (49)$$

can be written as the sum of volatility part and a correlation part. Since $\Sigma_t = D_t^{1/2} \Gamma_t D_t^{1/2}$ and D_t is diagonal, we have

$$\log |\Sigma_t| = \log |D_t| + \log |\Gamma_t| \quad (50)$$

$$(r_t - \mu_t)' \Sigma_t^{-1} (r_t - \mu_t) = (r_t - \mu_t)' (D_t^{1/2} \Gamma_t D_t^{1/2})^{-1} (r_t - \mu_t) \quad (51)$$

$$= u_t \Gamma_t^{-1} u_t + (r_t - \mu_t)' D_t^{-1} (r_t - \mu_t) - u_t' u_t \quad (52)$$

where $u_t = D_t^{-1/2} \varepsilon_t$ is vector of normalized innovations.

$$\log(L_T(\theta)) = -\frac{1}{2} \sum_{t=1}^T [n \log(2\pi) + \log |\Sigma_t| + (r_t - \mu_t)' \Sigma_t^{-1} (r_t - \mu_t)] \quad (53)$$

$$= -\frac{1}{2} \sum_{t=1}^T [n \log(2\pi) + \log |D_t| + \log |\Gamma_t| + u_t' \Gamma_t^{-1} u_t + (r_t - \mu_t)' D_t^{-1} (r_t - \mu_t) - u_t' u_t] \quad (54)$$

$$= -\frac{1}{2} \sum_{t=1}^T [n \log(2\pi) + \log |D_t| + (r_t - \mu_t)' D_t^{-1} (r_t - \mu_t)] \rightarrow \log(L_T^V(\theta)) \quad (55)$$

$$+ -\frac{1}{2} \sum_{t=1}^T [\log |\Gamma_t| + u_t' \Gamma_t^{-1} u_t - u_t' u_t] \rightarrow \log(L_T^C(\theta)). \quad (56)$$

Multivariate GARCH models – Estimation of DCC model

The last two terms of the second equality are clearly equal, but this expression allows the log-likelihood to be broken down in the two following terms:

$$\log(L_T(\theta_V, \theta_C)) = \log(L_T^V(\theta_V)) + \log(L_T^C(\theta_V, \theta_C)) \quad (57)$$

with

$$\log(L_T^V(\theta_V)) = -\frac{1}{2} \sum_{t=1}^T \left[n \log(2\pi) + \log |D_t| + (r_t - \mu_t)' D_t^{-1} (r_t - \mu_t) \right] \quad (58)$$

$$= -\sum_{i=1}^n \left[\frac{T}{2} \log(2\pi) + \frac{1}{2} \sum_{t=1}^T \left[\log(\sigma_{i,t}^2) + \left(\frac{r_{i,t} - \mu_{i,t}}{\sigma_{i,t}} \right)^2 \right] \right] \quad (59)$$

$$\log(L_T^C(\theta_V, \theta_C)) = -\frac{1}{2} \sum_{t=1}^T \left[\log |\Gamma_t| + u_t \Gamma_t^{-1} u_t - u_t' u_t \right] \quad (60)$$

$\log(L_T^V(\theta_V))$ is the sum of log-likelihoods of the individual GARCH equations.

Multivariate GARCH models – Estimation of DCC model

The second step consists in estimating the parameters pertaining to the correlation matrix, conditional on the parameters estimated in the first stage. (Since squared residuals are not dependent on correlation parameters, these parameters can be ignored for the estimation of the conditional volatility dynamics.)

The two-step estimation then relies on maximizing the log-likelihood as follows.

1. Estimate the volatility parameters through

$$\hat{\theta}_V \in \arg \max_{\{\theta_V\}} \log(L_T^V(\theta_V)) \quad (61)$$

2. Estimate the correlation parameters through

$$\hat{\theta}_C \in \arg \max_{\{\theta_C\}} \log(L_T^C(\hat{\theta}_V, \theta_C)) = -\frac{1}{2} \sum_{t=1}^T \left[\log |\Gamma_t| + \hat{u}_t \Gamma_t^{-1} \hat{u}_t - \hat{u}_t \hat{u}_t \right] \quad (62)$$

with $\hat{u}_{i,t} = (r_{i,t} - \hat{\mu}_{i,t}) / \hat{\sigma}_{i,t}$.

Multivariate GARCH models – Estimation of DCC model

The two-step estimator is consistent and asymptotically normal, with distribution

$$\sqrt{T}(\hat{\theta}_{TS} - \theta_0) \rightarrow N(0, A_0^{-1} B_0 (A_0^{-1})') \quad (63)$$

where

$$A_0 = -\frac{1}{T} E \sum_{t=1}^T \begin{bmatrix} \frac{\partial^2 \log(\ell_t^V(\theta_{V0}))}{\partial \theta_V \partial \theta_V'} & 0 \\ \frac{\partial^2 \log(\ell_t^C(\theta_0))}{\partial \theta_V \partial \theta_C'} & \frac{\partial^2 \log(\ell_t^C(\theta_0))}{\partial \theta_C \partial \theta_C'} \end{bmatrix} \quad (64)$$

$$B_0 = \frac{1}{T} E \sum_{t=1}^T \begin{bmatrix} \frac{\partial \log(\ell_t^V(\theta_{V0}))}{\partial \theta_V} \frac{\partial \log(\ell_t^V(\theta_{V0}))'}{\partial \theta_V} & \frac{\partial \log(\ell_t^C(\theta_0))}{\partial \theta_C} \frac{\partial \log(\ell_t^V(\theta_{V0}))'}{\partial \theta_V} \\ \frac{\partial \log(\ell_t^V(\theta_{V0}))}{\partial \theta_V} \frac{\partial \log(\ell_t^C(\theta_0))'}{\partial \theta_C} & \frac{\partial \log(\ell_t^C(\theta_0))}{\partial \theta_C} \frac{\partial \log(\ell_t^C(\theta_0))'}{\partial \theta_C} \end{bmatrix} \quad (65)$$

Due to the structure of A_0 , the asymptotic variances of the GARCH parameters θ_V for each series are the standard robust covariance matrix estimators. For the second stage parameters, however, the asymptotic variance involves all parameters.

Multivariate GARCH models – Illustration

Daily index returns from January 1980 to December 2018

	SP500-DAX		SP500-FT-SE	
	estimate	std err.	estimate	std err.
Diagonal BEKK model				
ω_{11}	0.1013	0.0114	0.0901	0.0124
ω_{12}	0.0512	0.0103	0.0519	0.0107
ω_{22}	0.1257	0.0140	0.0752	0.0152
a_{11}	0.2531	0.0157	0.2324	0.0161
a_{22}	0.2493	0.0147	0.1479	0.0116
b_{11}	0.9632	0.0045	0.9693	0.0044
b_{22}	0.9633	0.0038	0.9852	0.0029
log-lik.	-27913	-	-26494	-
Scalar BEKK model				
ω_{11}	0.1029	0.0087	0.0716	0.0129
ω_{12}	0.0504	0.0092	0.0431	0.0097
ω_{22}	0.1240	0.0127	0.0860	0.0146
a	0.2514	0.0114	0.1763	0.0133
b	0.9633	0.0031	0.9814	0.0034
log-lik.	-27913	-	-26565	-

Multivariate GARCH models – Illustration

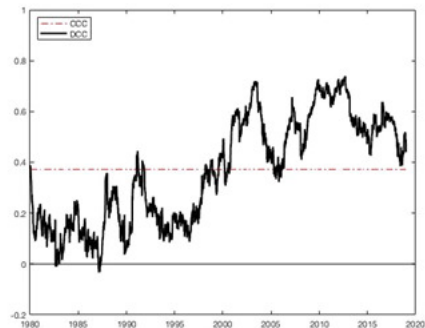
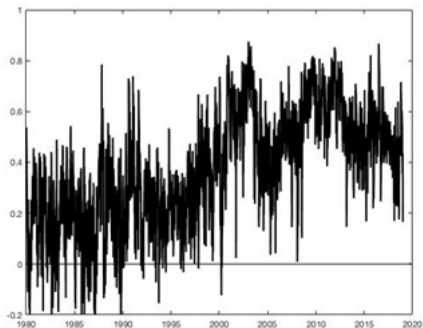
Daily index returns from January 1980 to December 2018

	SP500-DAX		SP500-FT-SE	
	estimate	std err.	estimate	std err.
Univariate GARCH models				
ω_1	0.0116	0.0030	0.0116	0.0030
a_1	0.0680	0.0083	0.0680	0.0083
b_1	0.9213	0.0098	0.9213	0.0098
ω_2	0.0272	0.0077	0.0073	0.0037
a_2	0.0940	0.0146	0.0220	0.0043
b_2	0.8912	0.0148	0.9714	0.0074
CCC model				
ρ	0.3725	0.0138	0.3628	0.0277
log-lik.	-28090	-	-26698	-
DCC model				
α	0.0082	0.0022	0.0051	0.0019
β	0.9913	0.0025	0.9948	0.0021
log-lik.	-27736	-	-26423	-

Multivariate GARCH models – Illustration

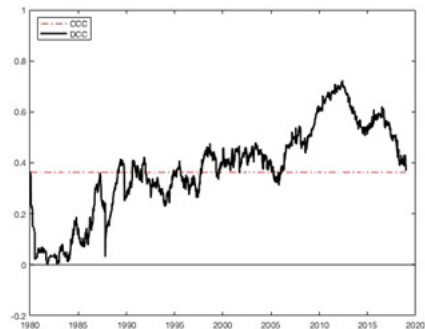
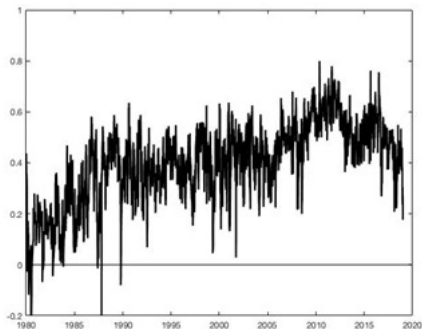
SP500 and DAX daily returns from January 1980 to Dec. 2018

BEKK model CCC and DCC GARCH models



Multivariate GARCH models – Illustration

SP500 and FTSE daily returns from January 1980 to Dec. 2018
BEKK model CCC and DCC GARCH models



Empirical betas from DCC

Why empirical betas belong in a DCC class

DCC is usually presented as a model of dynamic correlations. But in asset management, one of its most useful outputs is a time-varying covariance matrix.

If H_t denotes the conditional covariance matrix of returns, then conditional betas are immediate objects:

$$\beta_{i|m,t} = \frac{\text{Cov}_t(r_{i,t}, r_{m,t})}{\text{Var}_t(r_{m,t})}. \quad (66)$$

Engle's point is simple: once the covariance matrix is dynamic, empirical beta is also dynamic. It does not need to be estimated as a separate rolling regression.

The DCC beta formula

Let m be the benchmark or market asset. In the DCC model:

$$H_t = D_t R_t D_t, \quad (67)$$

where D_t contains conditional standard deviations and R_t is the conditional correlation matrix.

The beta of asset i to the benchmark m is:

$$\beta_{i|m,t} = \frac{h_{im,t}}{h_{mm,t}} = \rho_{im,t} \frac{\sigma_{i,t}}{\sigma_{m,t}}. \quad (68)$$

This decomposition is useful because beta can move because of correlation, relative volatility, or both.

A rolling OLS beta estimates:

$$r_{i,t} = \alpha_t + \beta_{i|m,t}^{OLS} r_{m,t} + u_{i,t} \quad (69)$$

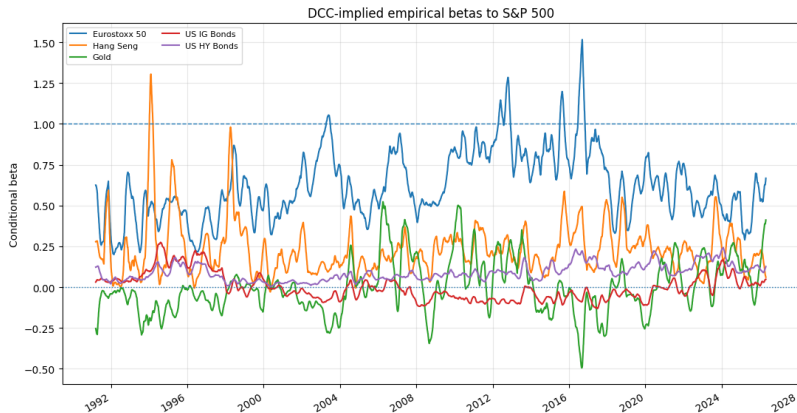
over a window chosen by the researcher.

A DCC-implied beta uses the conditional covariance matrix directly:

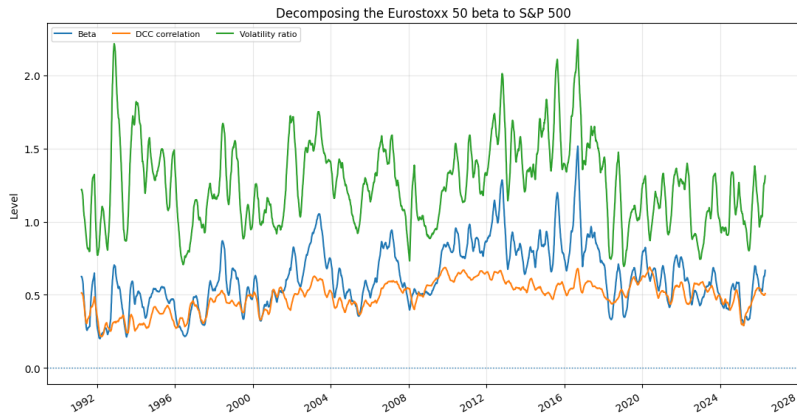
$$\beta_{i|m,t}^{DCC} = \frac{h_{im,t}}{h_{mm,t}}. \quad (70)$$

The advantage is conceptual: the same covariance model that drives risk, diversification and portfolio construction also drives beta exposures.

Empirical illustration: betas to S&P 500



What drives the beta?



For a portfolio with weights w_t , the conditional beta of the portfolio to asset j is:

$$\beta_{p|j,t} = \frac{\text{Cov}_t(r_{p,t}, r_{j,t})}{\text{Var}_t(r_{j,t})} = \frac{w_t' H_t e_j}{h_{jj,t}}. \quad (71)$$

This is different from risk contribution:

- risk contribution asks: which assets create portfolio volatility?
- empirical beta asks: how does the portfolio co-move with a given asset?

Both are computed from the same object: the conditional covariance matrix.

Multivariate distributions – Normal Mixture

A natural extension of the normal distribution is the normal mixture. The idea is to introduce randomness into the covariance matrix, via a positive mixing variable, W .

The random vector $Z = (Z_1, \dots, Z_n)$ is a multivariate normal variance mixture if

$$Z = \mu + \sqrt{W}AX \quad (72)$$

with:

- $X \sim N_n(0, I_n)$
- $W \geq 0$ is a non-negative, scalar valued r.v. independent of X
- μ is an $(n \times 1)$ vector and A is an $(n \times n)$ vector.

The distribution of Z conditional on a realization w of W is

$$Z|(W = w) \sim N_n(\mu, w\Sigma) \text{ with } \Sigma = AA'. \quad (73)$$

with

$$E[Z] = E[\mu + \sqrt{W}A X] = \mu + E[\sqrt{W}]A E[X] = \mu \quad (74)$$

$$\text{Var}[Z] = E[(\sqrt{W}A X)(\sqrt{W}A X)'] = E[W]A E[XX']A' = E[W]\Sigma \quad (75)$$

This class of distributions shows that lack of correlation does not necessarily imply independence of the components.

Let (Z_1, Z_2) have a normal mixture distribution with $A = I_2$ and $E[W] < \infty$ (i.e., no correlation, since $\text{cov}[Z] = E[W]AA' = E[W]I_2$). Then Z_1 and Z_2 are independent iff W is almost surely constant, i.e. (Z_1, Z_2) are normally distributed. We have:

$$\begin{aligned} E[Z_1 Z_2] &= E[W X_1 X_2] = E[W]E[X_1]E[X_2] \\ &\geq \left(E[\sqrt{W}]\right)^2 E[X_1]E[X_2] = E[Z_1]E[Z_2] \end{aligned} \tag{76}$$

where the equality holds only when W is constant.

Normal mixture distributions allow introducing non-linear dependence between the components.

Example: Let W be a r.v. with an inverse gamma distribution $Ig(\nu/2, \nu/2)$, or equivalently let $\nu/W \sim \chi^2(\nu)$. Then Z has a **multivariate t distribution** with ν degrees of freedom: $Z \sim t_n(\nu, \mu, \Sigma)$.

In this case, the multivariate t distribution is

$$f(z) = \frac{\Gamma\left(\frac{\nu+n}{2}\right)}{(\pi\nu)^{n/2} \Gamma\left(\frac{\nu}{2}\right) |\Sigma|^{1/2}} \left(1 + \frac{(z - \mu)' \Sigma^{-1} (z - \mu)}{\nu}\right)^{-\frac{\nu+n}{2}} \quad (77)$$

Remark 1: The variance of Z is not Σ . Indeed the mean of an inverse gamma $Ig(\alpha, \beta)$ is $\beta/(\alpha - 1)$, so that $E[W] = \nu/(\nu - 2)$ and therefore $\text{cov}[Z] = \frac{\nu}{\nu-2} \Sigma$.

Remark 2: This definition of the t distribution is not well appropriate for a conditional model, because the innovations Z have been preliminary standardized.

Multivariate distributions – Student t distribution

For a conditional (GARCH) model, we need to define a standardized version of the **t distribution** with ν degrees of freedom. This distribution should have zero means and unit individual variances. Such a distribution is defined as

$$f(z|\nu) = \frac{\Gamma\left(\frac{\nu+n}{2}\right)}{(\pi(\nu-2))^{n/2} \Gamma\left(\frac{\nu}{2}\right) |\Sigma|^{1/2}} \left(1 + \frac{z' \Sigma^{-1} z}{\nu-2}\right)^{-\frac{\nu+n}{2}} \quad (78)$$

where Σ is a correlation matrix.

In this case, each marginal distribution is symmetric with

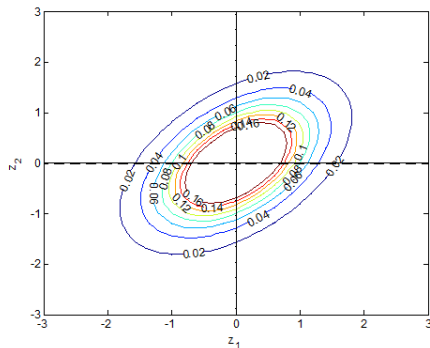
$$E[z_i] = 0, \quad (79)$$

$$V[z_i] = 1, \quad (80)$$

$$Sk[z_i] = 0 \quad (81)$$

$$Ku[z_i] = 3(\nu-2)/(\nu-4). \quad (82)$$

Contour plot of the Student t distribution with $\nu = 4$ and $\rho = 0.5$



Estimation

Now, we consider the estimation of this model for a vector of n returns $r_t = (r_{1,t}, \dots, r_{n,t})'$. Innovations z_t are defined by $z_t = \Sigma_t^{-1/2}(r_t - \mu_t)$, where μ_t denotes the conditional mean of r_t . Innovations are assumed to be distributed as a multivariate Student t distribution: $z_t | I_{t-1} \sim f(z_t | \nu)$.

The sample log-likelihood function of the multivariate GARCH model is written as

$$\log L_T(\theta, \eta) = -\frac{1}{2} \log |\Sigma_t(\theta)| + \sum_{t=1}^T \log \left(\underbrace{f(\Sigma_t(\theta)^{-1/2}(r_t - \mu_t(\theta))}_{\hat{z}_t} \mid \eta) \right) \quad (83)$$

where θ denotes the parameters pertaining to conditional mean and conditional volatility equations and $\eta = \nu$ denotes the degree of freedom.

Conclusion

Most of the time, estimating multivariate distributions is difficult for 2 main reasons:

- it is hard to find a distribution that describes the properties of several series simultaneously
- multivariate distributions are hard to estimate because they involve a lot of parameters.

Remark: in the case of the Student-t distribution, there is only one parameter to estimate in η but several parameters in θ .

That is why when the number of series increases, we usually take another approach, based on copulas.

A Couple Words on Copulas

In many situations where marginal distributions are not Gaussian, it is simply impossible to define a joint distribution.

This is the case when the two variables have **different marginal distributions** (for instance, a Student t variate and a Pareto variate). This is also the case for a large number of marginal distributions, for which a multivariate extension does not exist.

In such contexts, a solution is to use copula models. These functions have the property to relate two marginal distributions instead two series directly. Therefore, once margins have been estimated, no reference is made to their true functional form. **Copula models are able to relate any kind of margin.**

With copula models, it is possible to generate **non-linear dependence**. Therefore, in general, the Pearson's correlation is not the appropriate measure of dependence across variables.

A Couple Words on Copulas

Definition: A **bivariate copula** is a function $C : [0, 1] \times [0, 1] \rightarrow [0, 1]$ with the three following properties:

1. $C(u, v)$ is increasing in u and v .
2. $C(u, 0) = 0$, $C(u, 1) = u$, $C(0, v) = 0$, $C(1, v) = v$.
3. $\Pr[u_1 \leq U \leq u_2, v_1 \leq V \leq v_2] = C(u_2, v_2) - C(u_2, v_1) - C(u_1, v_2) + C(u_1, v_1) \geq 0$, $\forall u_1, v_1, u_2, v_2$ in $[0, 1]$ such that $0 \leq u_1 \leq u_2 \leq 1$ and $0 \leq v_1 \leq v_2 \leq 1$.

Property 1: When one marginal distribution is constant, the joint probability will increase provided that the other marginal distribution increases.

Property 2: If one margin has zero probability to occur, then it must be the same for the joint occurrence. Also, if on the contrary, one margin is certain to occur, then the probability of a joint occurrence is determined by the remaining margin probability.

Property 3: If u and v both increase, then the joint probability also increases. This property is, therefore, a multivariate extension of the condition that a cdf is increasing.

A Couple Words on Copulas

If we set $u = F(x)$ and $v = G(y)$, then $C(F(x), G(y))$ yields a description of the joint distribution of X and Y .

Theorem (Sklar, 1959): Let H be a joint distribution function of X and Y with marginal distributions F and G , respectively. Then,

- there exists a copula $C : [0, 1] \times [0, 1] \rightarrow [0, 1]$ such that, for all real numbers (x, y) ,

$$H(x, y) = C(F(x), G(y)) \quad (84)$$

Furthermore, if F and G are continuous, then C is unique.

- conversely, if C is a copula and F and G are univariate distribution functions, then $H(x, y) = C(F(x), G(y))$ is a joint distribution function with marginal distributions F and G .

The density of a copula is related to its cdf through the following relation

$$c(u, v) = \frac{\partial^2 C(u, v)}{\partial u \partial v}. \quad (85)$$

Similarly, the density h of the distribution H is defined by the relationship

$$h(x, y) = c(F(x), G(y)) \times f(x) \times g(y) \quad (86)$$

A Couple Words on Copulas

A first approach to the modeling of non-linear dependence consists in estimating the unrestricted joint density non-parametrically. This method has been proposed by Deheuvels (1979, 1981) to deduce a non-parametric estimate of the associated unrestricted copula.

The **empirical copula** is defined as

$$\hat{C}_T \left(\frac{t_1}{T}, \frac{t_2}{T} \right) = \frac{1}{T} \sum_{t=1}^T 1_{\{x_t \leq x_{(t_1)}, y_t \leq y_{(t_2)}\}} \text{ with } 1 \leq t_1, t_2 \leq T, \quad (87)$$

where $x_{(t)}$ denotes the order statistics, i.e., $x_{(1)} \leq \dots \leq x_{(T)}$ are ordered realizations.

The advantage of this approach is that it does not require any additional assumption on the non-linear dependence.

However, it suffers from the drawbacks of any non-parametric approach. In particular, the interpretation of the patterns of non-linear dependence is often complicated, and it is likely to provide inaccurate and erratic results, even in the bivariate case.

A Couple Words on Copulas

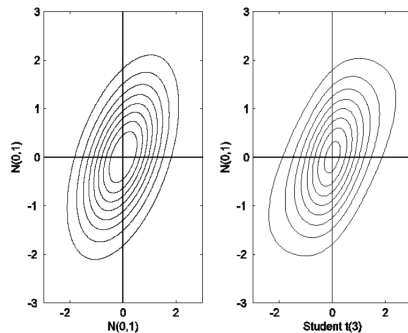
Then, there exists a whole library of parametric copulas, classified into families:

- The **elliptical** family: the Gaussian copula, the Student-t copula...
- **Archimedian copulas**: the Clayton copula (joint left tail dependency), the Rotated Clayton copula (upper tail dependence), the Gumbel copula (upper tail, rotated for lower tail).

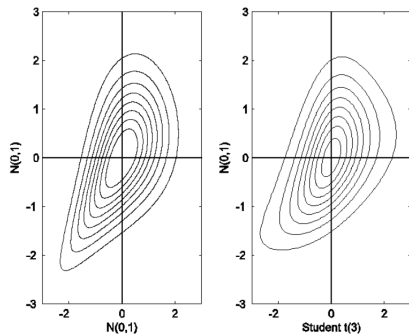
Remarks:

- Estimation done via maximum likelihood most of the time.
- The Elliptical family can have a dimension n when such a large scale copula is harder to obtain for Archimedian copulas.

Gaussian Copula - Contour Plot



Clayton Copula - Contour Plot



Empirical application: covariance
dynamics, ERC and betas

The notebook supplied with the class reproduces the empirical analysis.

The workflow is deliberately simple:

1. load multi-asset prices and carry forward missing observations,
2. compute daily log returns,
3. estimate rolling covariance matrices,
4. solve the ERC portfolio month by month,
5. compare weights, risk contributions and performance,
6. build a simple EWMA dynamic correlation filter to connect the application to DCC intuition,
7. compute DCC-implied empirical betas for assets and for the ERC portfolio.

Step 1: estimate rolling covariance matrices

At each monthly rebalancing date, estimate:

$$\hat{\Sigma}_t = 252 \times \frac{1}{m-1} \sum_{s=t-m+1}^t (r_s - \bar{r}_t)(r_s - \bar{r}_t)'. \quad (88)$$

In the notebook, $m = 504$ trading days, i.e. about two years.

This is not a GARCH covariance matrix. It is a transparent first empirical approximation. The econometric motivation for the rest of the class is that Σ_t can be modeled dynamically instead of estimated with a rolling window.

Step 2: solve the ERC problem

For each date t , solve:

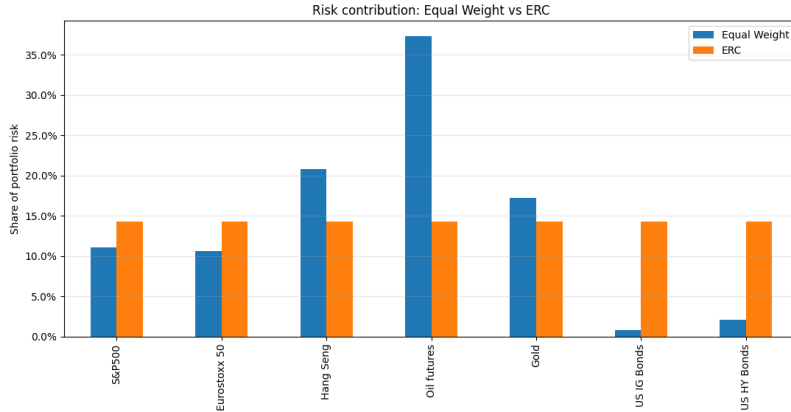
$$\min_{w_t} \sum_{i=1}^n \left(\frac{RC_{i,t}}{\sigma_{p,t}} - \frac{1}{n} \right)^2 \quad (89)$$

subject to:

$$\sum_{i=1}^n w_{i,t} = 1, \quad w_{i,t} \geq 0. \quad (90)$$

This gives a portfolio whose ex-ante risk is distributed equally across assets, according to the estimated covariance matrix.

Step 3: compare risk contributions



A full DCC model updates the correlation matrix using standardized shocks.

A simpler empirical approximation is the EWMA covariance recursion:

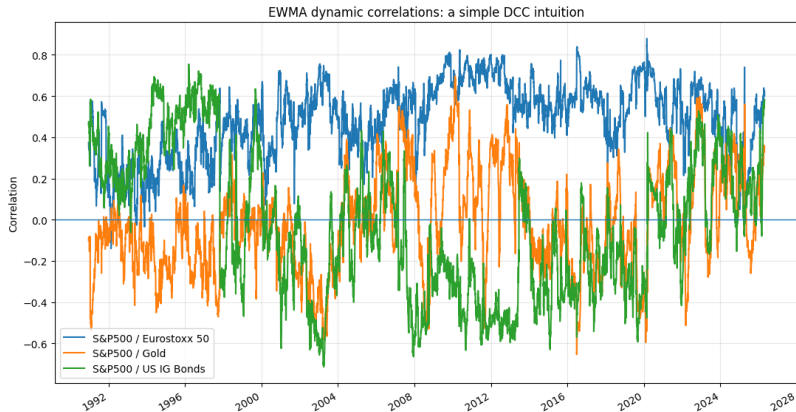
$$S_t = \lambda S_{t-1} + (1 - \lambda) r_{t-1} r'_{t-1}. \quad (91)$$

The implied correlation matrix is:

$$C_t = \text{diag}(S_t)^{-1/2} S_t \text{diag}(S_t)^{-1/2}. \quad (92)$$

This is not DCC estimation, but it is useful to visualize why dynamic correlations matter.

Dynamic correlations are state variables



Step 5: compute empirical betas from the DCC covariance

The notebook implements a transparent DCC-style filter:

$$Q_t = (1 - a - b)\bar{Q} + au_{t-1}u'_{t-1} + bQ_{t-1}, \quad R_t = \text{diag}(Q_t)^{-1/2}Q_t\text{diag}(Q_t)^{-1/2}. \quad (93)$$

Then it reconstructs:

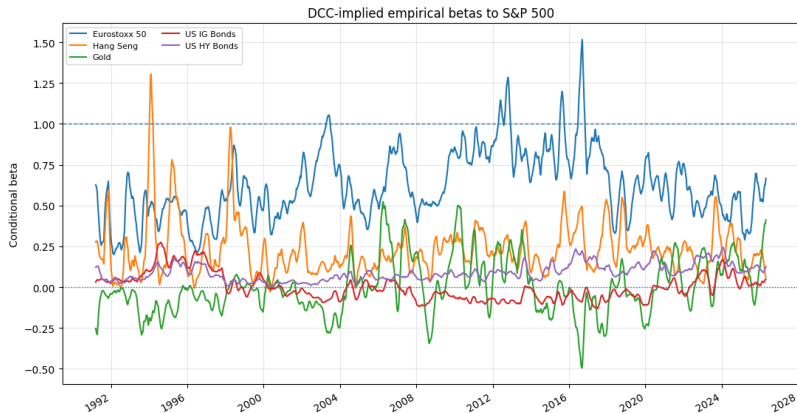
$$H_t = D_t R_t D_t \quad (94)$$

and computes betas as:

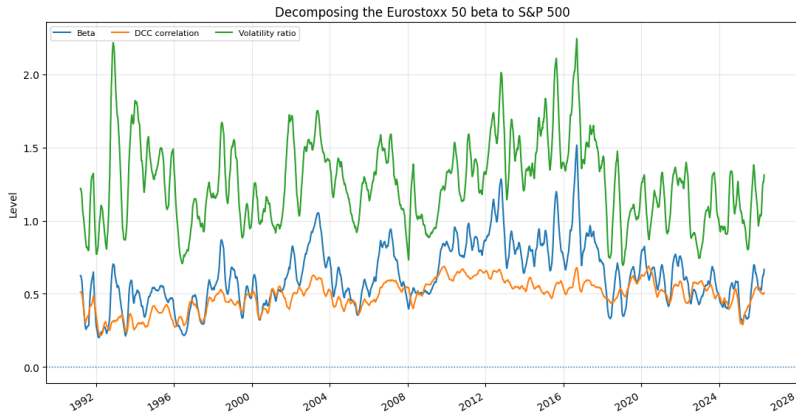
$$\beta_{i|m,t} = \frac{h_{im,t}}{h_{mm,t}}. \quad (95)$$

The objective is not industrial calibration. It is to make the bridge between DCC, covariance matrices and empirical beta visible.

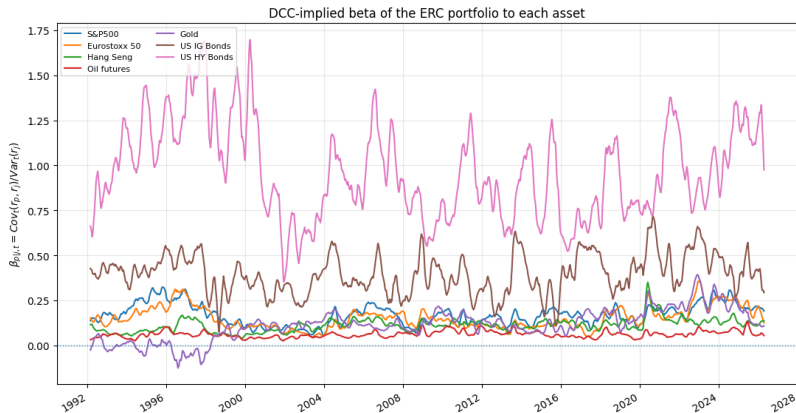
Application: DCC-implied empirical betas



Application: beta is correlation times relative volatility



Application: beta of the ERC portfolio to each asset



The ERC portfolio has equal ex-ante risk contributions, but it does not have equal betas to all assets.

This is expected, because:

$$RC_{i,t} = w_{i,t} \frac{(H_t w_t)_i}{\sigma_{p,t}} \quad \text{while} \quad \beta_{p|j,t} = \frac{w_t' H_t e_j}{h_{jj,t}}. \quad (96)$$

Risk contribution normalizes by portfolio volatility. Beta normalizes by the variance of the asset used as reference.

Use the notebook to answer seven questions:

1. How do ERC weights change if the covariance window is one year instead of two years?
2. Does ERC still reduce drawdown if commodities are excluded?
3. How different are inverse-volatility and ERC portfolios?
4. Which asset receives the highest risk contribution in the equal-weighted portfolio?
5. During which periods do equity-bond and equity-gold correlations change the most?
6. Change the benchmark for empirical betas from S&P 500 to Eurostoxx 50. Which betas are most unstable?
7. Compare the ERC portfolio beta to US investment-grade bonds and US high-yield bonds. What does it say about credit exposure?

What to remember

- Multivariate volatility modeling is about the full covariance matrix, not only variances.
- Correlations rise and fall over time, and this changes diversification.
- ERC is a practical way to see why Σ_t matters for asset allocation.
- DCC-implied empirical betas come directly from H_t : $\beta_{i|m,t} = h_{im,t}/h_{mm,t}$.
- DCC is attractive because it separates variance dynamics from correlation dynamics.
- Non-Gaussian dependence and copulas matter when tail co-movements are stronger than linear correlation suggests.